A call for sharing phenotypic antimicrobial susceptibility data alongside bacterial genome sequence data in public archives

Genomics is widely used in the surveillance and investigation of antimicrobial resistance (AMR), with standard practice to deposit bacterial whole-genome sequence data in public archives to support research and public health. Here, we call for open sharing of matched phenotypic antimicrobial susceptibility data alongside sequence data to support equitable data-driven development and validation of new technologies to address AMR, and highlight key principles and resources to facilitate such sharing.

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Introduction

Bacterial whole-genome sequencing (WGS) is vital to the surveillance and investigation of antimicrobial resistance (AMR) across the research, clinical, and public health domains[1]. It is standard practice to share bacterial sequence data, where legally and ethically permissible, through public archives of the International Nucleotide Sequence Database Collaboration (INSDC), including the National Center for Biotechnology Information (NCBI), the European Molecular Biology Laboratory's European Bioinformatics Institute (EMBL-EBI), and others. Depositing WGS in INSDC databases is a requirement of many funders, publishers, and surveillance programmes, including PulseNet or GenomeTrakr. These WGS data support a variety of activities, including research (e.g., on pathogen biology, epidemiology and evolution), methods development, and contextualisation of sequence data generated for clinical and public health purposes.

Several emerging genomic applications that aim to understand and address AMR require WGS data to be matched with other data types including antimicrobial susceptibility testing (AST) data. In particular, to harness the power of artificial intelligence (AI) and other data-driven approaches, it will be essential to expand the availability of large, high-quality, genetically and geographically diverse, open-access WGS/AST datasets, especially for priority AMR pathogens[2,3]. However, as of June 2026, AST data is available for fewer than 2.5% of bacterial pathogen genomes in NCBI (https://www.ncbi.nlm.nih.gov/pathogens/ast) or EMBL-EBI (https://www.ebi.ac.uk/amr/), even after the CABBAGE project extracted AST data from other public databases and from literature and deposited these in EMBL-EBI[4]. The availability of matched WGS/AST data can facilitate innovation, benchmarking and validation of new tools (such as sequence-based diagnostics and phenotype prediction[5]), and identification of novel AMR mechanisms[6]. Notably, most applications of sequencing to the investigation of AMR (including isolate or metagenomic-based sequencing for clinical, public health, or environmental surveillance) rely on databases of known AMR determinants[7], such as the Comprehensive Antibiotic Resistance Database (CARD)[8] or NCBI's Reference Gene Catalog[9]. However, these AMR marker databases include many genetic variants that have not been fully experimentally or clinically validated, making the clinical interpretation of AMR genotypic profiles challenging[10]. Improving the quality and interpretability of AMR marker databases, and advancing our collective knowledge of AMR mechanisms, would be greatly enhanced by increased availability of matched WGS/AST data across a broad range of bacterial pathogens. Here, we call on the global health and scientific community to normalise the practice of depositing accompanying AST data when submitting WGS data to public archives. We discuss key practical considerations for sharing AST data, including data standards and formats, and highlight new tools to support preparing AST data for sharing.

Why share

As highlighted previously[11,12], the use of genomics and computational analysis, including machine learning and other AI approaches, depends critically on the availability of large-scale, high-quality, standardised phenotypic AST data linked to WGS data, for representative isolate collections including susceptible and resistant strains[13]. This has a wide range of potential applications, including (i) assessing the impact of a genetic variant on phenotype; (ii) identifying novel resistance determinants and mechanisms[6]; (iii) building validated predictors that could be used for molecular diagnostics or detection (whether sequencing-based, such as metagenomic sequencing direct from specimens, with or without pathogen enrichment; or designing suitable targets for rapid non-sequencing-based molecular assays such as qPCR or CRISPR-based diagnostics); and (iv) similar applications in surveillance across a range of contexts, such as the hospital built environment, hospital or municipal wastewater, natural or agricultural environments, animal, and food systems[11]. Therefore, from a cost-benefit and policy perspective, sharing existing contextual data such as phenotypic AST takes costs relatively little in terms of staff time, but increases the value and utility of bacterial WGS data, with consequent implications for improving the cost-effectiveness of sequencing programmes[14–17]. Notably, the availability of matched WGS/AST data is skewed towards high-income countries, with fewer data available from lower-resourced settings. This could lead to genotypic-based predictions or interpretations that fail to generalise across geographies, novel AMR mechanisms arising in specific geographies may be missed, and pathogens of importance in low-resource settings may be underrepresented (e.g., despite the phenotypic surveillance efforts of the Global Task Force on Cholera Control, there is there is hardly any *Vibrio cholerae* AST data in NCBI and none in EMBL-EBI at the time of writing).

Publicly accessible matched WGS/AST datasets enable more researchers to focus on developing new analytical methods using a common, globally representative resource. Open data is also essential for transparent benchmarking of emerging methods for antimicrobial phenotype prediction, reducing selection bias and allowing developers to evaluate and compare model performance using common reference datasets[18,19]. Additionally, publicly available data could facilitate making evidence linking AMR genotype to phenotype more accessible to users without species-specific expertise, advanced genomics knowledge or computational capacity to analyse raw WGS data. For example, new tools could be built that allow microbiologists to rapidly query the evidence supporting a specific phenotypic interpretation for a genetic variant reported by a WGS pipeline or molecular assay in an unfamiliar pathogen. Additionally, public health practitioners could use these data to investigate whether changes in AMR prevalence are driven by specific bacterial lineages[20] and/or molecular mechanisms[21].

What to share

When sharing AST data, the essential elements to include are (i) details of the assay method or platform, including model or version of any instrument or standard used; (ii) name of the tested antimicrobial (using standardised ontologies[8]); and (iii) the numerical assay measurements, e.g., minimum inhibitory concentration (MIC) or inhibition zone diameter (see **Table 1**). The numerical measurements should include the sign ('=', '>', '<=') to convey whether the measured value was observed directly (e.g., MIC = 2 mg/L), or was inferred to be above or below the range assessed by the specific assay (e.g., MIC <=2 mg/L). It can be useful to share contextual metadata on the isolates themselves; however, the granularity of such data needs to be carefully managed with due consideration to privacy and confidentiality laws, and the inability to share detailed metadata should not preclude public deposition of sequence and AST data (metadata are discussed further below).

Sharing numerical assay results is critical because categorical interpretations such as 'S' (susceptible) and 'R' (resistant), or above *vs* below an epidemiological cut-off value (ECOFF), are not fixed (and do not exist for all bug-drug combinations). Clinical breakpoints differ between interpretive standards, with notable differences between the two most commonly referenced organisations for interpreting AST data from human clinical isolates: Clinical and Laboratory Standards Institute (CLSI) and European Committee on Antimicrobial Susceptibility Testing (EUCAST)[22,23]. These differ in their definitions of the 'I' category, which drives some differences in breakpoints[24]. Within a given standard, breakpoints change over time, which can lead to major interpretive errors if datasets collected over multiple years are categorised using two different resistance thresholds. Even within the same interpretive standard and version, different breakpoints may be set for different clinical syndromes or formulations (e.g., meningitis *vs* other infections); EUCAST and CLSI also have distinct standards for use in veterinary settings. Notably, the dilution range tested varies across instrument panels, meaning that capped values such as ≥4 and ≥32 mg/L may both be compatible with the same true MIC value if above the tested range. Recording the testing platform and reagent format can help downstream users interpret such values appropriately. Additionally, S/I/R categorization may be inferred from reporting rules rather than interpreted from assay measurements (e.g., EUCAST recommends all *Klebsiella pneumoniae* be reported as 'R' for ampicillin regardless of the MIC). These adjustments are often not distinguishable from interpretations derived directly from the assay measurement, making it difficult for downstream users to assess the provenance of categorical results. Importantly, there are also regional and sectoral differences in the standards used, which can introduce geographical and ecological bias when aggregating interpreted phenotypes from different sources.

Sharing the numerical measurement, together with details of the method used to produce it, means that data aggregated from different sources can be re-interpreted with consistent breakpoints. For example, since 2012 the CLSI breakpoints for ciprofloxacin resistance in *Salmonella enterica* serovar Typhi are S ≤0.06 mg/L and R ≥1 mg/L, resulting in three categories (S/I/R), whereas the EUCAST breakpoint (since 2014) is R >0.06 mg/L, dividing the population into two categories only (S/R). To compare CLSI-based data from North America with EUCAST-based data from European reference laboratories, it is necessary to combine the numerical MIC data and then re-interpret these using a single set of breakpoints. The “S/I/R” interpretations themselves are not directly comparable when generated using the two different standards, as they give the false impression that isolates analysed in Europe have a much higher prevalence of resistance than those tested in North American laboratories, whereas this is primarily a consequence of the differences in reporting standards between laboratories and the underlying MIC distributions are actually quite similar (**Figure 1**)[25]. Details of the method are critical, and are the main indicator of the quality or precision of public AST data. Sharing method details allows data users to assess technical bias, restrict their analysis to data from specific platforms or methods if necessary, and to assess the generalisability of predictive algorithms across methods.

There are some potential exceptions to the requirement to share numerical assay measurements. Firstly, if only categorical AST data are available, it may still be valuable to share in this form, and EMBL-EBI allows this (NCBI does not). For example, where a laboratory using disk diffusion in resource-limited settings only recorded the S/I/R and not the measurement, uploading categorical AST data could still be highly beneficial for gaining global representation in INSDC repositories. However, it is essential to provide full details of the assay methods used and the standard (with version) applied to reach that interpretation, so that downstream users can determine which interpretive breakpoints were utilised and thus which datasets are suitable for inclusion in a given analysis (similar to tracking the version of a genotypic reference database used to infer AMR genes and mutations). Secondly, AST data may be generated using assays that do not produce MIC or disk inhibition zone diameter measures. For example, for *Mycobacterium tuberculosis*, S/R calls may be made by assessing a single critical concentration, in which case it is essential to share details of the media and drug concentration used. Other standardised clinical assays include the D-test (for inducible clindamycin resistance in Gram-positive bacteria) or Double Disk Synergy Test (to detect β-lactamase activity in Gram-negative bacteria). In some clinical reporting systems, these results may be used to update the interpretation to ‘R’ regardless of MIC or disk inhibition zone diameter measures (similar to inferred results based on reporting guidelines as noted above), and/or they may be reported in a separate field. There is currently no standard way to share such data; however, we recommend including separate fields to indicate the additional assay result. It would also be valuable to develop standardised fields for conveying more explicit assay information where available, e.g. antimicrobial concentration used in disk diffusion assays, and quality control information such as details of replicates or calibration standards.

	Proportion R, interpreted using local standard	Proportion R, re-interpreted with consistent standard	
		EUCAST (R >0.06 mg/L)	CLSI (R >=1 mg/L)
UKHSA (n=782)	n=69, 94.4% (EUCAST)	n=69, 94.4%	n=738, 8.8%
US CDC (n=720)	n=93, 12.9% (CLSI)	n=553, 76.8%	n=93, 12.9%

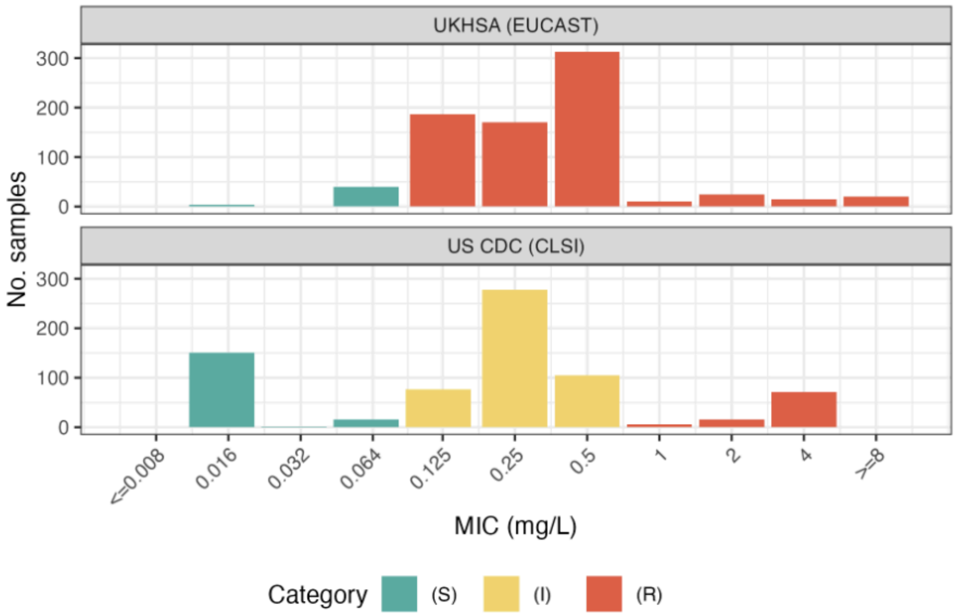

Figure 1. Importance of sharing numerical assay measures rather than categorical interpretations. Comparison of *S. enterica* serovar Typhi isolates categorised as resistant to ciprofloxacin, as reported by the United Kingdom Health Security Agency (UKHSA, n=782, years 2016-2020) *vs* those reported by the United States Centers for Disease Control and Prevention (US CDC, n=720, years 2016-2019) (data sourced from an aggregated dataset[25]). UKHSA uses the European Committee on Antimicrobial Susceptibility Testing (EUCAST) standard to interpret numerical assay measures (minimum inhibitory concentration, MIC), resulting in 94.4% of routine surveillance isolates being categorised as R (resistant). US CDC uses the US-based Clinical & Laboratory Standards Institute (CLSI) standard to interpret MIC values, resulting in just 12.9% of routine surveillance isolates being categorised as R. However the distribution of MIC values shows a modest difference (as shown in the bottom panel), whereas the very large difference in apparent resistance rate is mostly due to the difference in breakpoints used by the EUCAST *vs* CLSI interpretive standards. Re-interpreting the MIC values of both national datasets using the same interpretive guidelines, the differences in resistance rate are much more modest (e.g., 8.8% in the UK *vs* 12.9% in the US, using the CLSI standard, see upper panel).

Where to share

Scientific data should be shared according to FAIR principles (Findable, Accessible, Interoperable, Reusable)[26]. To maximise the findability and accessibility of matched WGS/AST data, both data types should be deposited in the same database and be linked by a common sample identifier. As INSDC is already the standard location for depositing WGS data, we advocate depositing AST data there as well, specifically within NCBI or EMBL-EBI, which provide mechanisms to deposit, search, and retrieve AST data linked to bacterial sequence data.

Depositing WGS/AST data in INSDC also ensures their interoperability and reusability. Sequence data are mirrored across EMBL-EBI, NCBI and other INSDC databases, and individual isolates are assigned a unique identifier known as the BioSample accession number[27]. AST data can be submitted to EMBL-EBI or NCBI using the same BioSample accession as the sequence data, at the time of sequence deposition or retrospectively, providing a clear and permanent link between sequence and phenotype (although AST data is not yet shared between INSDC databases). Notably, under the biological data object model used by EMBL-EBI and NCBI[28], the same BioSample accession can be linked to one or more sequencing experiment records relating to that sample (which provide information on the experimental approach and sequencing platform used to generate a sequence dataset), as well as to the resulting sequence data. A single BioSample may be linked to multiple sequencing experiments and associated with uniquely accessioned sequence data files that may include different types of sequence data (e.g., WGS short reads, WGS long reads, genome assemblies, transcriptomics data). In addition, EMBL-EBI and NCBI each provide enhanced AMR genotyping data for certain pathogens, which are linked to the BioSample (see https://ebi.ac.uk/amr, https://www.ncbi.nlm.nih.gov/pathogens/antimicrobial-resistance/).

Both databases have minimal contextual metadata requirements for BioSample records associated with pathogen genome data submissions (see https://www.ncbi.nlm.nih.gov/biosample/docs/packages/, https://www.ebi.ac.uk/ena/browser/view/ERC000028), which include date of isolation (e.g., year), location (e.g., country or latitude and longitude coordinates), source, and (where relevant) host and disease/health status. More extensive contextual metadata may also be submitted, including clinical outcome where available[29]. It is helpful to include structured provenance metadata to help downstream users distinguish clinical isolates from laboratory-selected mutants, genetically modified strains, repeated patient isolates, or outbreak-enriched collections that may bias genotype-phenotype analyses, and skew prevalence which in turn skews prediction metrics. Importantly, by sharing AST data in NCBI or EMBL-EBI, these phenotype data are automatically linked (via BioSample accession) to all publicly available genetic data for the same isolate, including derived AMR genotypes and raw WGS data, as well as basic source information, in a publicly accessible and searchable platform.

How to share

NCBI and EMBL-EBI each require AST data to be submitted in a specific format (see **Table 1**). These have broadly corresponding core elements, addressing the essential points outlined above (i.e., details of the susceptibility testing assay used and the resulting numerical assay measurements). These fields map closely to those included in data collection templates used by regional and global public health surveillance programmes, including the World Health Organization (WHO) Global AMR Surveillance System (GLASS)[30]. Thus, these AST data formats are generally interoperable, although some conversion is required.

Table 1. Comparison of AST submission data standards used by NCBI and EMBL-EBI. Comparison of fields in the data templates used for submitting AST data to the public sequence archives NCBI or EMBL-EBI, as of June 2026. Note that both include fields for antibiotic, the interpreted resistance phenotype (SIR), the reference standard or guidelines used (e.g. CLSI, EUCAST), numerical assay measurement values from which these interpretations were made, including measurement signs (i.e. ==, >, >=, <, <=) and units, and the typing platform (e.g., Sensititre, MicroScan, Vitek, Phoenix, etc.; this is optional as data can be generated without automated platforms). The NCBI template also has optional fields to specify the platform's vendor and its reagent format. The EMBL-EBI template has an additional optional field 'breakpoint version' to indicate the version of the testing standard (e.g., EUCAST 2015). * Mandatory fields. '-' indicates the template has no corresponding field.

NCBI	EMBL-EBI	Examples		
*biosample_accession	*accession	SAMN31087099	SAMN26304659	SAMN13836992
*antibiotic	*antibioticName	ciprofloxacin	ciprofloxacin	gentamicin
*testing_standard	*astStandard	EUCAST	CLSI	EUCAST
*resistance_phenotype	*resistancePhenotype	resistant	resistant	susceptible
*measurement_sign	*measurementSign	==	>	==
*measurement	*measurement	4	2	17
*measurement_units	*measurementUnits	mg/L	mg/L	mm
*laboratory_typing_method	*laboratoryTypingMethod	broth dilution	broth dilution	disk diffusion
laboratory_typing_platform	platform	Sensititre	Phoenix	
vendor	-	Trek	Becton Dickinson	
laboratory_typing_method_ version_or_reagent	-	96-Well Plate	96-Well Plate	
-	breakpointVersion			

To facilitate formatting AST data for submission to NCBI and EMBL-EBI, our AMRgen R package[31] includes functions to import data from different sources (including WHONET formats, or files exported from several automated AST instruments) and export it into either a tabular format suitable for submission to NCBI (https://www.ncbi.nlm.nih.gov/biosample/docs/antibiogram/) or JSON files suitable for submission to EMBL-EBI (https://www.ebi.ac.uk/amr/amr_submission_guide/). The AMRgen Converter web application (http://apps.amrverse.org/amrgenconverter/) provides an interactive online web interface to these functions (**Figure 2**), enabling users to upload input files of AST data (exported from automated AST instruments or the WHONET software recommended by WHO GLASS[32]), and optionally a file mapping isolate identifiers to BioSample accessions, and download files suitable for submitting the AST data to NCBI or EMBL-EBI.

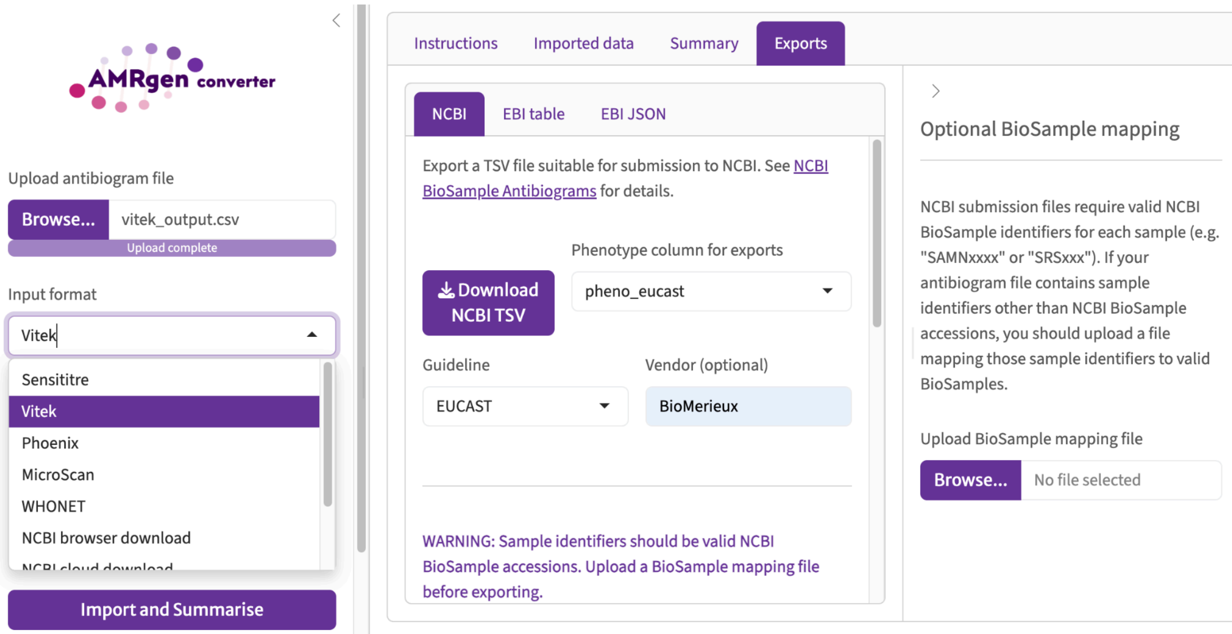

Figure 2. Formatting AST data for submission to the NCBI or EMBL-EBI BioSample databases using the AMRgen Converter web application. Primary input to the AMRgen Converter web application is an AST data file exported directly from an automated commercial AST instrument (e.g., Sensititre, Vitek, Phoenix, MicroScan) or AST data in WHONET format. The user must select an input file to upload, then click 'Import and Summarise'. The user may then view the uploaded AST data table (under the 'Imported data' tab), or a summary of its contents (under the 'Summary' tab), for review before exporting the data. Clicking on the 'Exports' tab reveals a download window, where the user must choose the tab corresponding to the desired download format (e.g. 'NCBI'), and provide any additional information needed for the submission template (e.g. the guideline used, such as 'EUCAST'). NCBI/EMBL-EBI submissions must label samples with a BioSample accession, a unique identifier that is also used to label the corresponding sequence data. As the uploaded AST data would typically use a different sample identifier (e.g. a laboratory code), users may also upload an optional 'BioSample mapping' file providing the links between the uploaded sample identifiers and valid BioSample accessions to be added to the exported files. The app is written in R Shiny utilising the import and export functions of the AMRgen package for R[31], it is suitable for users unfamiliar with R and is available online at http://apps.amrverse.org/amrgenconverter/. Alternatively, the app code can be downloaded from https://github.com/AMRverse/AMRgenconverter and run locally, or experienced R users may prefer to use the underlying AMRgen functions directly. Examples of how to extract data from selected instruments are available at https://esgem-amr.amrrules.org/export-ast.

NCBI also supports an *M. tuberculosis*-specific antibiogram format, suitable for sharing AST results based on a single critical concentration (https://www.ncbi.nlm.nih.gov/biosample/docs/antibiogram-myco/). Notably, there is currently no provision in any NCBI or EMBL-EBI template to convey the results of alternative assay data (e.g., D-test), nor to indicate that an S/I/R call was made on some basis other than interpretation against a breakpoint (e.g., based on reporting rules, expected phenotypes, or D-test). This omission is presumably because such information is not considered important for surveillance purposes; however, it is important for biological understanding, and so data standards should be updated to incorporate such data as a priority to support data sharing and scientific discovery. Addition of a field to explicitly indicate whether a reference method was followed (e.g. ISO 20776-1 broth microdilution) would also be very helpful, to allow easy identification of reference-quality data.

Conclusion

Public bacterial genome archives are rapidly becoming the foundation for AMR surveillance, diagnostics and discovery. Their value will remain fundamentally limited unless phenotype accompanies genotype. Routine deposition of matched AST data should therefore become standard practice wherever bacterial genome sequences are shared, and we conclude with a call to action for the global scientific community to support this (**Table 2**).

Firstly, whenever bacterial sequence data are deposited in NCBI or EMBL-EBI, the corresponding AST data should also be submitted and linked through the BioSample accession, with retrospective deposition of AST data encouraged for previously archived genomes or those where AST data is not yet available at the time of sequence data submission. Secondly, researchers should actively encourage and support colleagues and collaborators to do so, and consider using recommended co-creation frameworks to boost representative datasets from currently underrepresented geographies and lower-resource settings. To incentivise this, users of public data should ensure they cite datasets appropriately, including individual BioProjects and associated publications wherever possible, so that data generators can receive appropriate credit for the effort involved in sharing their data. Thirdly, journals should mandate public deposition of AST data that is essential to the reproducibility of a combined WGS/AST study; and peer reviewers should flag to authors and editors that both the sequence and AST data should be submitted to the public archives, linked by BioSample accession, rather than providing AST data in supplementary files or non-INSDC archives[4,12]. Where deposition is not possible because of legal, ethical, or governance restrictions, authors should provide a clear explanation and share the maximum permissible metadata. Reviewers also have a role to play in ensuring that public data is appropriately cited and credited when used in publications. Finally, funders supporting the generation of matched AST and WGS data should mandate and resource public deposition of the data, which should be documented in data management plans. Funding support for digitization of laboratory information management, especially in low-resource settings, would also facilitate data sharing, amongst other benefits.

Following these steps will benefit the global AMR community, not only those who directly access this matched WGS/AST data for their own research and activities, but also anyone who uses AMR genotype databases – be it for research, clinical application, surveillance, or as a comparator for alternative approaches. Public availability of this data will enhance database quality, improve representativeness, and enable robust external validation across the field.

Table 2. Call for action. Actions that different stakeholders can take to improve the quality, quantity, diversity, and representativeness of matched WGS/AST data available in public archives.

Stakeholder	Actions
Data generators	**Submit AST data to NCBI/EMBL-EBI** ● In future, whenever submitting new sequences ● Now, to supplement previously deposited sequences ● Complete as many fields as possible to make the methodology clear and maximise re-usability **Plan for submission of sequence and AST data in new projects** ● Include this in data management plans, and plan for requesting any necessary approvals or resources **Encourage and support colleagues and collaborators to do the same, but be sensitive to individual challenges and concerns**
Data users	**Give appropriate credit whenever using public data** ● Report and acknowledge the database used (e.g., NCBI AST browser, EBI AMR Portal) ● Cite individual studies wherever feasible (these can be identified via the BioProject accession field, or the PubMed ID field in data sourced from the EBI AMR portal)
Journals	**Mandate public deposition of both WGS and matched AST data that is essential to the reproducibility of a study** ● Adopt a policy that where co-analysis of WGS and AST data are central to a study, both the WGS and AST data must be submitted to INSDC databases and linked by BioSample
Peer reviewers	**Indicate to authors and editors that sample-level WGS and matched AST data should be publicly deposited** ● Where authors have submitted AST data in supplementary files, advise that this data should also be deposited in the public archives, and linked to the sequence data by BioSample accession ● Where authors have co-analysed WGS and AST data but not shared one or both data types, advise that both should be deposited in NCBI or EMBL-EBI, linked by BioSample accession ● Where authors have used public data, check for appropriate citation of the source database and individual studies
Funders	**Mandate INSDC deposition of both WGS and matched AST data that is generated using research grant funds** ● Require that data management plans include a commitment to this, on an appropriate time scale, or a justification for why it cannot be done ● Fund costs associated with data management and sharing (e.g., data entry or software for digitisation of manual assay results, data manager to support data cleaning and formatting)

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Competing Interests

The authors declare no competing interests.

Data availability

The example dataset analysed during the current study, to generate Figure 1, is available in the GitHub repository, (https://github.com/typhoidgenomics/TyphoidGenomicsConsortiumMykrobe, DOI: https://doi.org/10.5281/zenodo.17117922) in file 'Supplementary_Table_4.csv'.

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